

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #5: due Thursday, Mar. 12th, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.
While you can check your answers with Wolfram Alpha, remember that you're going to have to do
integrals by hand on the exam!

1. (25 points.) Consider the following PDF:

$$f_X(x) = \begin{cases} \frac{3}{35}x^2 & x \in [-2, 3] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (8 points.) Compute $E(X)$.
- (b) (8 points.) Compute $\text{Var}(X)$.
- (c) (9 points.) Compute $E(\cos(X^3))$.

2. (30 points.)

Let $Y = 7X^2 - 1$, where X is the same random variable from Problem 1.

- (a) (10 points.) Compute $E(Y)$, based only on your answers from Problem 1.
- (b) (20 points.) Compute the CDF of Y . Note this is a little tricky; think about how the values of X map to values of Y . Drawing a picture is a good idea. Don't forget to specify the ranges of values where each "piece" of the CDF is valid.

3. (20 points.) Let X be a random variable representing the length of an episode of the new Scrubs reboot, in minutes. Initially we don't know anything about the PDF of X , other than that it is symmetric about its mean.

- (a) (7 points.) Suppose we learn that $E(X) = 22$. Compute the Markov bound on $P(X \geq 25)$.
- (b) (8 points.) We now further learn that $\text{Var}(X) = 2$ (note this is the variance, not the standard deviation). Compute the Chebyshev bound on $P(|X - 22| \geq 3)$, and then divide this by two to get an estimate of $P(X \geq 25)$ since we know the distribution is symmetric.
- (c) (5 points.) Finally, we learn the complete distribution of X : it's actually a Gaussian with the mean and variance given above. Compute the actual value of $P(X \geq 25)$. How does it compare to the bounds above?

4. (25 points.) A company makes waterslides that have a semi-circular cross-section with radius R , where R is modeled as a uniform random variable on $[1.5, 2]$ meters.
- (a) (15 points.) The length of the waterslide is $L = 64$ meters. Let V be the (random) volume of the waterslide if it's full of water (remember, this is equal to its length times its cross-sectional area, $\frac{1}{2}\pi R^2 L$). Determine the PDF of V .
- (b) (10 points.) The weight of lost belongings W in kilograms recovered from a waterslide is related to its volume V by the equation $W = \left(\frac{V}{\pi}\right)^{\frac{1}{2}}$. Compute the PDF of W . (Hint: this is easier than it looks if you determine what kind of random variable W is.)

As above, don't forget to specify the ranges of values where each piece of the PDF is valid.